Precision For A Qubit Operation and Failure of Achieving Quantum Supremacy

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Abstract

M. Dyakonov [IEEE Spectrum, 2019] asked: what precision is required for manipulating a quantum gate, and what role is the notion of ENERGY playing in quantum computing theory. We answer the questions by investigating the Quantum Fourier Transformation (QFT). We show that the precision for a qubit operation is no less than 4.72694×10^{-36}, in accordance with the energy of a free particle. Quantum entanglement is indispensable to quantum supremacy. The present quantum computers have failed to instantiate quantum entanglement up to our expectation, due to inexact qubit operations. Quantum transition incurred during the transition of space-time continuum covertly changes qubits and corrupts quantum supremacy.

Keywords: Planck constant, Quantum Fourier Transformation, Bell's inequality, EPR thought experiment, quantum entanglement, quantum transition.

1. Introduction

Quantum Fourier Transformation (QFT) plays a key role in quantum computation theory [1], which needs to apply the rotation gates

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & \exp(-2\pi i/2^k) \end{bmatrix}, k = 2, 3, \cdots, n$$

to some qubits, where n is the number of qubits used for QFT. The physical manipulation of gates $R_k, k > \ell$, for some global parameter ℓ, is infeasible due to the failure of controlling extremely weak energy. In 1994, Coppersmith [2] introduced the Approximate QFT (AQFT), and claimed that the matrix entries of AQFT differ from those of QFT by a multiplicative factor of $\exp(i\epsilon)$. In 2004, Fowler and Hollenberg [3, 4] suggested a new scaling technique for QFT, called Banded QFT (BQFT). Its basic idea is to replace any gate $R_k, k > \ell$, by

$$R_{\ell,\xi} = \begin{bmatrix} 1 & 0 \\ 0 & \exp(-2\pi i\xi/2^\ell) \end{bmatrix}, \quad \text{where } \xi \geq 1.$$

In 2013, Nam and Blümel [5, 6, 7] discussed the scaling laws for BQFT.

The Copenhagen interpretation for quantum mechanics says that a measurement causes an instantaneous collapse of wave function and the state after the collapse is random. Besides, two quantum systems can be combined into a composite system by using tensor product. Einstein, et al. [8] insisted that quantum mechanics was incomplete and proposed a thought experiment to refute it.

In 1964, J. Bell [9] mathematically formulated EPR experiment and constructed an inequality,

$$|E(\vec{a}, \vec{b}) - E(\vec{a}, \vec{c})| \leq 1 + E(\vec{b}, \vec{c}) \tag{1}$$

which involves three vectors in a Hilbert space. In 1969, Clauser, et al. [10] generalized the Bell's formulation to the case involving four vectors, i.e.,

$$|E(\vec{a},\vec{b}) - E(\vec{a},\vec{b}') + E(\vec{a}',\vec{b}') + E(\vec{a}',\vec{b})| \leq 2 \tag{2}$$

In the past decades, many experiments [11, 12, 13, 14, 15] have been designed to test the Bell's inequality or CHSH inequality [10, 16, 17]. Today, the inequalities are of great importance to quantum computation and quantum computers [15, 18, 19]. Quantum entanglement is considered indispensable to quantum supremacy. The present quantum computers failed to instantiate quantum entanglement. What's the reason for failures of these quantum computers? Is it possible to manufacture a quantum computer with supremacy up to our expectation in the future?

M. Dyakonov asked a crucial question what precision was required for manipulating a quantum gate. He also commented that the notion of ENERGY, of primordial importance in physics, both classical and quantum, has absolutely no role in quantum computing theory [20].

In this paper, we show that the precision for a qubit operation is no less than 4.72694×10^{-36} in accordance with the energy of a free particle. It is the first time to obtain such a precision and clarify the role of energy in quantum computing theory. We also argue that the failure of instantiating quantum entanglement originates from quantum state transition, which is incurred during the transition of space-time continuum. We believe, quantum transition is a basic principle for any quantum state, like that the mass and acceleration distort and interact with space-time continuum. We find this principle can be used to tackle some questions in quantum mechanics, especially the EPR experiment and the failures of quantum computers.

2. Preliminaries

The state of a qubit is described by a 2-dimensional vector $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$, where α and β are complex numbers such that $|\alpha|^2 + |\beta|^2 = 1$. The basic two quantum states corresponding to the two states of a classical bit are defined by bit 0 $\leftrightarrow |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, bit 1 $\leftrightarrow |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The basic single-qubit operations include $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}, S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$, where H is called Hadamard gate. Clearly, $H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), H|1\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Given two separate qubits, the corresponding two-qubit state is given by the Kronecker product of vectors. For example, $\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha \begin{bmatrix} \gamma \\ \delta \end{bmatrix} \\ \beta \begin{bmatrix} \gamma \\ \delta \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{bmatrix}$. The basis for two-qubit states consists of

$$\text{string } 00 \leftrightarrow |00\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \text{string } 01 \leftrightarrow |01\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix},$$

$$\text{string } 10 \leftrightarrow |10\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \text{string } 11 \leftrightarrow |11\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

A unitary transformation on n qubits is a matrix U of size $2^n \times 2^n$. Sometimes, two-qubit gates can be described by the tensor product of some single-qubit gates. Not all two-qubit gates can be written as the tensor product of single-qubit gates. Such a gate is called an *entangling gate*, for example, the CNOT gate. The gates H, T and CNOT form a universal gate set because any general unitary transformation can be broken into a series of two qubit rotations.

The only way to change qubits without measuring is to apply a unitary operation. Quantum computations can be created by designing unitary operations in sequence, each of which is composed of smaller operations.

3. Quantum Fourier Transformation

Let n be the number of qubits used for QFT, $N = 2^n$, and $\omega = \exp(-2\pi i/N)$. The QFT for n-qubits is described by the matrix

$$\mathrm{QFT}_N = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \cdots & \omega^{(N-1)(N-1)} \end{bmatrix}$$

The QFT can be decomposed and represented by Kronecker product and binary fraction. Let $x_1x_2\cdots x_n$ be a binary string. Its binary fraction is expressed as

$$0.x_1x_2\cdots x_n = \frac{x_1}{2} + \frac{x_2}{2^2} + \cdots + \frac{x_n}{2^n}$$

The QFT performed on state $|x_1x_2\cdots x_n\rangle$ can be represented by

$$\begin{aligned} |x_1x_2\cdots x_n\rangle \xrightarrow{\mathrm{QFT}} & \frac{1}{\sqrt{2^n}}[|0\rangle + e^{-2\pi i 0.x_n}|1\rangle] \otimes [|0\rangle + e^{-2\pi i 0.x_{n-1}x_n}|1\rangle] \\ & \otimes \cdots \otimes [|0\rangle + e^{-2\pi i 0.x_1x_2\cdots x_n}|1\rangle] \end{aligned} \tag{3}$$

and further translated into the below quantum circuit (Fig.1, Ref.[21]).

4. Uncontrollable gates

QFT needs a very huge transform size $N = 2^{1024}$ if Shor algorithm is used to fact RSA-1024. Is it possible to run QFT with such a transform size? The answer could be discouraging due to the difficulty to physically implement the basic operators. Actually, the Planck constant is $h = 6.626 \times 10^{-34}$ joule second. The Planck length is 1.62×10^{-35} meters, the smallest possible length. The Planck time is 5.391247×10^{-44} seconds, an incredibly small interval of time. But now, we have

$$R_{1024} = \begin{bmatrix} 1 & 0 \\ 0 & \exp(\frac{-2\pi i}{2^{1024}}) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \cos(\pi/2^{1023}) - i\sin(\pi/2^{1023}) \end{bmatrix},$$

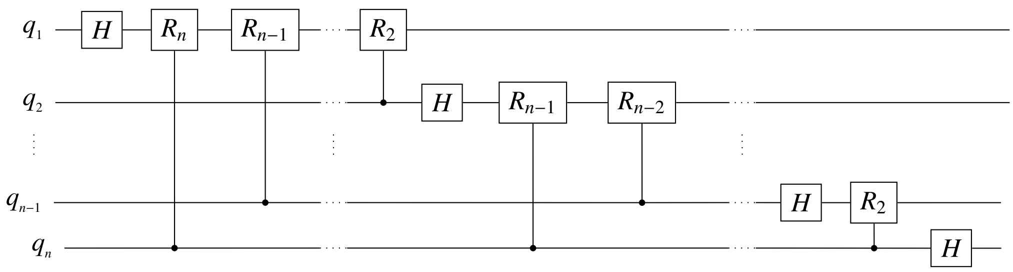

Figure 1: QFT circuit for n qubits

where

$$\sin(\pi/2^{1023}) \approx 3.49514 \times 10^{-308},$$

is an extremely tiny number.

5. Precision for a qubit operation

Notice that the energy of a free particle with wave function Ae^{ikx} is

$$E = \frac{h^2k^2}{8\pi^2 m},$$

where m is the mass of this particle, x is the distance between an observer and this particle. In the considered scenario, $k = \frac{2\pi}{2^n x}$, the approximate value of its energy is

$$\frac{h^2}{2^{2n+1}x^2 m}.$$

The rest mass of an electron is $m \approx 9.109 \times 10^{-31}$ kg. Hence, the energy of this electron is about $\frac{7.457\times10^{-654}}{x^2}$ (joul), if $n = 1024$.

Since the Planck length is 1.62×10^{-35} meters, we have the approximation

$$\frac{7.457 \times 10^{-654}}{2.6244 \times 10^{-70}} \approx 2.84 \times 10^{-584} \text{ (joule)}$$

which is an extremely tiny number. Clearly, the operator R_{1024} acted on an electron (or other quantum particle) cannot generate any physical quantity change, such as energy and frequency.

Practically, the lowest frequency of radio waves is only 3 Hertz, with the energy 1.9878×10^{-33} joule. In view of this fact and taking into account the roundoff and truncation errors, we think, 2^{120} could be an upper bound to the transform size of QFT, corresponding to the extreme energy value 5.19724×10^{-40} (joule). This means only the operators $R_2, R_3, \cdots, R_{120}$, could be manipulated by the current quantum technology. The above argument shows the energy dependency of QFT, and the precision for a qubit operation is no less than

$$\sin(\pi/2^{119}) \approx 4.72694 \times 10^{-36}.$$

6. Good news and bad news about quantum computers

There were so much news about quantum computers. For example,

- In 2009, Google demonstrated quantum computer image search [22].
- In 2011, a quantum computer was sold to high-profile client [23].
- In 2013, Google and NASA team was up to use quantum computer [24].
- In 2014, the world's first buyers' guide for quantum computers was available [25].
- In 2015, quantum technology was set to hit the streets within two years [26].
- In 2016, Google's plan for quantum computer supremacy was revealed [27].
- In 2017, IBM has built two new quantum computers with 20 and 50 qubits [28].
- In 2018, Google has built 72-qubit chip [29].
- In 2019, Google has achieved quantum supremacy [30].
- In 2019, Google's quantum supremacy algorithm has found its first practical use [31].
- In 2020, Honeywell announced the world's highest performing quantum computer [32].
- In 2020, IonQ unveiled world's most powerful quantum computer [33].
- In 2020, D-Wave claimed it has the world's most powerful quantum computer [34].
- In 2021, IBM unveiled breakthrough 127-qubit quantum processor [35].
- In 2022, IBM unveiled world's largest quantum computer at 433 qubits [36].
- In 2023, IBM quantum computer beat a supercomputer in a head-to-head test [37].
- In 2024, Google's claim of quantum supremacy has been completely smashed [38].
- In 2024, D-Wave said quantum computers can solve otherwise impossible tasks [39].
- In 2025, Microsoft unveiled Majorana 1, the world's first quantum processor powered by topological qubits [40].
- In 2025, quantum computers were on track to solve knotty mathematical problems [41].
- In 2025, Microsoft's quantum computer was hit with criticism at key physics meeting [42].
- In 2025, doubts were casted over D-Wave's claim of quantum computer supremacy [43].

We also have noticed the news, "Google launched $5m prize to find actual uses for quantum computers" [44]. As so far, the present quantum computers have not yet actually demonstrated their supremacy.

7. Quantum transition incurred during the transition of space-time continuum

It is well known that the mass and acceleration distort and interact with space-time continuum. How about the effect of transition of space-time continuum? This issue has never been investigated, although there were the terms of quantum decay [45] and quantum dissipation [46], which refer to the loss of energy and coherence in a quantum system due to interactions with its environment. The new term, quantum transition, depends only on the transition of space-time continuum, without any interactions.

Let $\psi(\boldsymbol{x}, t)$ be the wave function of a qubit at the position $\boldsymbol{x}$ and time t. Then its new wave function $\psi(\boldsymbol{x}', t')$ at the position $\boldsymbol{x}'$ and time t' is represented as

$$\psi(\boldsymbol{x}', t') = e^{i(\frac{|\boldsymbol{x}'-\boldsymbol{x}|}{\mu} + \frac{|t'-t|}{\nu})}\psi(\boldsymbol{x}, t) \tag{4}$$

where $\mu = 1.616 \times 10^{-35}$ meters (Planck length), and $\nu = 5.391 \times 10^{-44}$ seconds (Planck time). In our opinion, the transition of space-time continuum always incurs quantum transition such that any quantum state cannot be strictly maintained.

If the transition principle is true, the interpretation for EPR thought experiment will be compatible with uncertainty principle, collapse theory, general relativity theory, and tensor product. We also find the transition principle can be used to explain why so many quantum computers in the past years failed to put into practice.

8. The core questions in EPR experiment revisited

As we know, Einstein firstly investigated the issue of quantum entanglement. The core questions raised in the EPR experiment consist of:

Q1. Is the measuring result certain or uncertain? If it is certain then it can be represented as a scalar. Otherwise, it is represented as a vector. Lots of practical testimonies confirm that the Heisenberg uncertainty principle is generally acceptable.

Q2. Is the quantum state transition instantaneous? If true, it is incompatible with the general relativity theory. Many researchers [47, 48, 49, 50, 51] have endeavored to fill the gap.

Q3. Is tensor product the unique candidate for describing a composite system? This question is rarely discussed. We would like to stress that if tensor product is not indispensable to quantum mechanics, the contradiction in EPR experiment will be eliminated, without rejecting uncertainty principle and collapse theory. In this case, there does not exist the strictly entangled pair of spins.

Disproving the tensor product for a quantum composite system seems hopeless and futile. Quantum transition, alternatively, could be a good choice for the EPR experiment. Namely, both two states of entangled spins are changed due to the incurred quantum transition. The two singletons are not strictly entangled after they are separated.

9. Practices are speaking out

The research of quantum computer in the past decades was mainly driven by quantum entanglement theory. Very recently, Google and XPRIZE have launched $5m prize to find actual uses for quantum computers [44]. This prize could calm some quantum entanglement and quantum computer fanatics down. Should we believe those experiments conducted in laboratories in colleges and universities, or the quantum machines produced by D-wave, Google, and IBM? Only practices are speaking out. If quantum transition principle is true for any quantum system, then the failures of quantum computers would be explainable. We think, the modulated quantum states inside these quantum computers do not accord with our prediction just owing to quantum transition and inexact qubit operations. The quantum transition principle can also be used to explain why so many quantum computers failed to put into practice, i.e., they failed to instantiate quantum entanglement due to the incurred quantum transition, which is covertly changing qubits.

How to confirm quantum transition? How to counteract the effect of quantum transition for qubits so as to instantiate quantum entanglement up to our expectation? Both could be essential to produce future quantum computers for actual uses.

10. Conclusion

We show that the Planck constant limits the transform size of QFT. From the practical point of view, the implementation of QFT over 120 qubits is infeasible. If more qubits are needed, QFT should be replaced by AQFT. It is the first time to show the explicit dependency of energy in quantum computing theory, and the precision for a qubit operation. We also find the present quantum computers failed to instantiate quantum entanglement up to our expectation. The quantum state of qubits cannot be strictly maintained due to the incurred quantum transition. We think, the study on principle of quantum transition for qubits would be useful for producing future quantum computers with true quantum supremacy.

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